

ANTHONY TY MARLER

Winnsboro, Texas 75494 · (903) 474-3647 · tymarler@gmail.com

Senior Application Architect · Inventor · Energy, Climate-Tech & AI-Assisted Systems

PROFESSIONAL SUMMARY

Results-oriented technical strategist and application architect who builds mission-critical software and data-driven systems for operational efficiency, regulatory compliance, and business outcomes. Named inventor on two issued U.S. patents. Combines AWS cloud, .NET, SQL, and systems integration with generative AI (Cursor, Claude, ChatGPT) and deep expertise in energy, mining, manufacturing, and environmental systems. Leads cross-functional teams through the full project lifecycle.

SELECTED HIGHLIGHTS

- **AI-assisted architecture** — Apply Cursor, Claude, and ChatGPT to design, code, document, and troubleshoot energy and industrial systems, pairing LLM workflows with architectural ownership and domain validation.
- **Vistra** — Designed a data-driven process engine for dynamic tasks; mine planning and monitoring software that measures productivity by shift; and a cloud emissions platform that unified diverse systems for credit trading, accounting for **\$60 million** in profit in the first two months.
- **Compaq** — Invented and developed a patented software-delivery system that eliminated more than **70% of computer manufacturing time**, plus a post-sales configuration method that increased inventory flexibility.
- **Texas Built Biofuels** — Invented a process-intensified biodiesel method that reduced reaction time from **4 hours to 4 minutes** (60x throughput; ~40% lower energy cost per gallon; ASTM-quality fuel).
- **Automated Response Information Systems** — Designed and implemented a hardware-based voice-recognition system for hands-free data collection in harsh environments (early applied AI / speech recognition).

CORE COMPETENCIES

Application Architecture · Generative AI (Cursor, Claude, ChatGPT) · Systems Integration · AWS Cloud & Chef · .NET / C# / SQL · Agile / Scrum · Cross-Functional Leadership · Root-Cause Analysis · TCO Analysis · Data Modeling & Database Design · Emissions Compliance & Carbon Credit Trading · Mining Operations Software · GIS · SharePoint · OSISOFT PI · Tableau

PROFESSIONAL EXPERIENCE

Independent | Texas

Mar 2025 – Present

Application Architect, AI-Assisted Systems

- Apply Cursor, Claude, and ChatGPT across architecture, coding, root-cause analysis, and documentation to accelerate delivery while retaining design ownership and domain review.
- Use AI-assisted development to prototype and refine cloud and application solutions (AWS, .NET, SQL) for energy, climate-tech, and industrial operations.

Vistra Corporate Services Company | Texas

Dec 2019 – Feb 2025

Senior Analyst / Architect, Generation

- Designed and developed a data-driven process engine to manage dynamic tasks, providing automated steps to drive the process to completion.
- Designed and developed mine planning and monitoring software to calculate productivity and measure results by shift.
- Designed and developed a cloud-based AWS platform that ingested SCADA, PLC, and third-party emissions data into a normalized database for credit trading; accounted for \$60 million in profit in the first two months, cut trading latency from weeks to hours, and reduced compliance reporting time by 80%.
- Led web applications supporting mining, fossil generation, battery, and solar operations, including Tableau dashboards for traders and compliance officers.

Accenture | Texas

Aug 2016 – Dec 2019

Business Application Manager

- Led day-to-day maintenance, enhancement, and operations for the Fossil Applications portfolio serving Luminant generation and mining.
- Directed development across AWS, Chef, .NET, C#, C++, PHP, SQL Server, MySQL, Tableau, Formotus, TDMobile, StackVision, and NetDAHS.
- Owned troubleshooting and RCA for production issues across ROTT, Skaief, SmartProcedures, GIS, Mobility, Safety Index, Proact, Digital Signage, Qdabra, SharePoint, PI, and land management.
- Led a cloud modernization that aligned DevOps, QA, and engineering and delivered three weeks ahead of schedule. Mentored engineers on AWS and Chef; a junior developer became a lead contributor within six months.

Senior Application Architect

- Architected and delivered Fossil Generation applications covering MSHA training, emissions compliance, procedures, GIS, safety, mobility, digital signage, and SharePoint.
- Provided architectural solutions for new builds and rebuilds to improve reliability and functionality, including TCO analysis of existing and future-state environments.
- Served as SME for mining operations, mining software systems, data modeling, and database design; introduced mobility (Formotus, TDMobile), OSIsoft PI, land management, and application monitoring.

ADDITIONAL EXPERIENCE

Texas Built Company, LLC | Winnsboro, TX

Feb 1998 – Feb 2008

Owner. Invented a process-intensified biodiesel reactor that reduced reaction time from four hours to four minutes, enabling continuous rather than batch production, about 60x throughput in the same footprint, and roughly 40% lower energy cost per gallon while maintaining ASTM fuel quality. Patent pending: Texas Built process intensified biofuels reactor technologies.

Samsung / AST Research

Mar 1997 – Feb 1998

Principal Engineer / Worldwide Software Systems Architect. Worldwide software systems architecture for PC manufacturing and configuration platforms.

Compaq Computer Corporation | Houston, TX

Oct 1993 – Mar 1997

Senior Systems Engineer. Invented and developed a patented software-delivery system that eliminated more than 70% of the manufacture time of their computers. Also invented a post-sales configuration method that increased inventory flexibility. Named inventor on U.S. Patents 6,202,070 and 5,974,567.

Automated Response Information Systems, Inc.

Oct 1991 – Oct 1993

Owner. Designed and implemented a hardware-based voice-recognition system for hands-free data collection in harsh environments (early applied AI / speech recognition).

Stanford Telecommunications, Inc.

Jul 1989 – Aug 1991

Systems Engineer. Worked on a team to develop planning and control software for the defense communication satellite network.

Elcom and Associates

Feb 1987 – Jul 1989

Field Engineer. Field engineering and technical support for communications and electronics systems.

U.S. Army

Feb 1984 – Feb 1987

Strategic Microwave System Repair Technician. Maintained strategic microwave communications; completed Army electronics, microwave repair, and cryptographic training.

PATENTS

- U.S. 6,202,070** (2001) — Computer manufacturing system architecture with enhanced software distribution functions (Compaq; named inventor).
- U.S. 5,974,567** (1999) — Ghost partition / runtime-selectable file system (Compaq; named inventor).
- Pending** — Texas Built process intensified biofuels reactor technologies.

TECHNICAL SKILLS

AI & developer tools: Cursor, Claude, ChatGPT; AI-assisted architecture, coding, RCA, and documentation

Languages & platforms: C# / .NET, VB.NET, C++ / C++.NET, C, PHP, Java, SQL, AJAX, Chef, Intel Assembly; prior: Ada, FORTRAN, BASIC, VMS/DCL, PRO*C, FOCUS, PAL, Silverlight

Cloud, data & monitoring: AWS, Amazon RDS, Chef, APIs, OSIsoft PI, MS SQL Server, MySQL, Oracle, DB2, Tableau, SCADA/PLC integration, BAM application monitoring, StackVision, NetDAHS

Applications & tools: Visual Studio, SharePoint, ESRI GIS, Formotus, TDMobile, Qdabra, SmartProcedures, Visio, MS Project, LabVIEW

Domains: Fossil generation, mining operations, emissions compliance and carbon credit trading, MSHA training, behavior-based safety, land management, mobility, digital signage, computer manufacturing software delivery, biofuels process intensification, AI-assisted industrial systems

EDUCATION & PROFESSIONAL DEVELOPMENT

Computer Science coursework and Certificate in Ada, Colorado Technical College. Advanced C++ and Advanced Java, Compaq. DEC VAX/VMS System Management. U.S. Army Basic Electronics, Strategic Microwave System Repair, and Cryptographic courses.